THE GENERAL NEUTRALITY THEOREM

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Abstract. The collision invariant S_ℓ satisfies the reflection identity $S(a) + S(m-a) = -1$ on $(\mathbb{Z}/m\mathbb{Z})^\times$. The companion paper proved neutrality at $q = 3$: a specific mod-3 class has mean $-1/2$. This paper proves the general version: for every odd prime q not dividing b, the reflection fixes exactly one residue class modulo q, and that class has mean $-1/2$ on the combined CRT period modulo mq.

These constraints can be imposed simultaneously on the combined CRT period for finitely many primes. By Mertens' third theorem, the fraction of lifted units that avoid neutrality at every prime $q \leq Q$ tends to zero as $Q \to \infty$.

The centered prime harmonic sum $F^\circ(s) = \sum \hat{S}^\circ(\chi)\, P(s,\chi)$ decomposes over odd characters. Computation of the products $\hat{S}^\circ(\chi) \cdot P(s,\chi)$ reveals that they have mixed signs: at $s = 1/2$ in base 10, thirteen of twenty are positive and seven are negative. The positive terms outweigh the negative in magnitude, not in count. This pattern persists across all prime bases tested up to $b = 37$, with the positive fraction drifting toward 50% as the number of characters grows, while the net sum remains positive.

The data therefore points toward a collective phenomenon. Convergence of F° would constrain the joint zero distribution of the odd L-functions modulo m, not each L-function individually.

1. Introduction

The collision invariant S_ℓ is a function on $(\mathbb{Z}/m\mathbb{Z})^\times$ with $m = b^{\ell+1}$, satisfying the reflection identity [3]

$$S(a) + S(m-a) = -1.$$

The companion paper [4] proved convergence of the centered prime harmonic sum $F^\circ(s)$ at $s = 1$ and identified the neutrality theorem at $q = 3$ as the mechanism controlling penetration below $s = 1$.

This paper extends the neutrality theorem to all odd primes and examines the sign structure of the character decomposition. The results separate the behavior of F° from the simplest positivity route to the

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Generalized Riemann Hypothesis: the mixed-sign computations indicate that convergence of F° need not be a test of each L-function individually, but rather a collective constraint on the family.

2. The General Neutrality Theorem

Definition 1. Let q be an odd prime with $q \nmid b$, and let $m = b^{\ell+1}$. For a residue class $k \bmod q$, define the lifted class

$$\mathcal{U}_{m,q}(k) = \{1 \le a \le mq : \gcd(a, m) = 1,\ a \equiv k \pmod{q}\}.$$

Its lifted mean is

$$\overline{S}_k^{(q)} = \frac{1}{\phi(m)} \sum_{a \in \mathcal{U}_{m,q}(k)} S_\ell(a \bmod m).$$

By the Chinese remainder theorem, each lifted class contains exactly $\phi(m)$ elements.

Theorem 2 (General neutrality)**.** *Let q be an odd prime with $q \nmid b$, and let $m = b^{\ell+1}$. Define*

$$k^* \equiv m \cdot \frac{q+1}{2} \pmod{q},$$

the unique solution to $2k \equiv m \pmod{q}$. Then:

1. *The reflection $a \mapsto m - a$ maps the class $\mathcal{U}_{m,q}(k^*)$ to itself modulo mq.*
2. *The lifted mean over this class equals $-1/2$:*

$$\overline{S}_{k^*}^{(q)} = -\frac{1}{2}.$$

Proof. The reflection sends class k to class $m - k \pmod{q}$. The fixed class satisfies $k \equiv m - k$, i.e., $2k \equiv m \pmod{q}$. Since q is odd, 2 is invertible and the solution k^* is unique.

The reflection lifts to an involution on $\mathcal{U}_{m,q}(k^*)$ modulo mq: if $a \in \mathcal{U}_{m,q}(k^*)$, let $a^\sharp$ be the unique representative in $[1, mq]$ satisfying

$$a^\sharp \equiv m - a \pmod{mq}.$$

Then $a^\sharp \equiv m - k^* \equiv k^* \pmod{q}$, and $a^\sharp \bmod m \equiv -a \bmod m$, so $a^\sharp$ is again coprime to m. Thus $a^\sharp \in \mathcal{U}_{m,q}(k^*)$.

Since S_ℓ is a function modulo m, the reflection identity gives

$$S_\ell(a \bmod m) + S_\ell(a^\sharp \bmod m) = -1.$$

Summing over the involution pairs and dividing by $|\mathcal{U}_{m,q}(k^*)| = \phi(m)$ yields $\overline{S}_{k^*}^{(q)} = -1/2$. $\square$

Corollary 3 (Paired classes)**.** *The remaining $q-1$ classes split into $(q-1)/2$ pairs $\{k,\, m-k \bmod q\}$ swapped by reflection. Their lifted means satisfy*

$$\overline{S}_k^{(q)} + \overline{S}_{m-k}^{(q)} = -1.$$

Corollary 4 (Simultaneous neutrality)**.** *For any finite set of odd primes $q \nmid b$, the neutrality constraints from Theorem 2 hold simultaneously on the combined CRT period modulo $m \prod q$.*

3. The Neutral Fraction

Fix a finite set $\mathcal{Q}$ of odd primes not dividing b, and set

$$M_{\mathcal{Q}} = \prod_{q \in \mathcal{Q}} q.$$

Consider the lifted unit period

$$\mathcal{U}_{m,\mathcal{Q}} = \{1 \le a \le m M_{\mathcal{Q}} : \gcd(a, m) = 1\}.$$

For each $q \in \mathcal{Q}$, let k_q^* denote the neutral class from Theorem 2. The fraction of lifted units that avoid neutrality at every prime in $\mathcal{Q}$ is exactly

$$\prod_{q \in \mathcal{Q}} \frac{q-1}{q}.$$

Indeed, by the Chinese remainder theorem, the conditions $a \equiv k_q^* \pmod{q}$ are independent across $q \in \mathcal{Q}$. For each fixed q, exactly one of the q residue classes is neutral, so the avoiding fraction is $(q-1)/q$, and the product is exact on the combined CRT period. By Mertens' third theorem, this product tends to zero as $\mathcal{Q}$ exhausts the odd primes not dividing b.

Remark 5. Each constraint fixes a class mean, not individual values. The collision invariant is not determined by neutrality. But it is far from generic: a random function lifted to the CRT period would satisfy none of these constraints.

4. The Sign Structure

The antisymmetry theorem [4] gives

$$F^\circ(s) = \sum_{\substack{\chi \bmod m \\ \chi(-1)=-1}} \hat{S}^\circ(\chi)\, P(s, \chi),$$

a sum over non-trivial odd characters. Each product $\hat{S}^\circ(\chi) \cdot P(s, \chi)$ is a complex number whose real part contributes to $F^\circ(s)$. If all these real

b	q	k^*	$\sum S$	count	$2\sum$+count
5	3	2	−3	6	0
5	7	2	−2	4	0
7	3	2	−7	14	0
7	5	2	−4	8	0
10	3	2	−6	12	0
10	7	1	−3	6	0
13	3	2	−26	52	0
13	5	2	−16	32	0
13	7	1	−11	22	0

TABLE 1. The identity $2\sum S + \text{count} = 0$ (mean $= -1/2$) holds exactly for all tested bases $b \leq 20$ and primes $q \leq 19$: 117/117 cases.

parts were non-negative, convergence of F° at $s = 1/2$ would require each $P(s, \chi)$ to converge individually, equivalent to the Generalized Riemann Hypothesis for the odd L-functions modulo m.

s	positive	negative	$\sum_+$	$\sum_-$	net
1.0	13	7	+0.47	−0.39	+0.08
0.8	15	5	+0.28	−0.10	+0.19
0.6	13	7	+0.75	−0.31	+0.44
0.5	13	7	+1.46	−0.73	+0.73

TABLE 2. Sign of $\mathrm{Re}(\hat{S}^\circ(\chi) \cdot P(s, \chi))$ across the 20 odd characters modulo 100. Base 10, lag 1, 348,488 primes. The products have mixed signs at every s.

The positivity condition fails. Seven of twenty products are negative at $s = 0.5$ in base 10. The net sum is positive because the positive terms have larger magnitude, not because they are more numerous.

5. Universality of the Sign Structure

Two patterns emerge across all bases tested.

First, the positive fraction drifts toward 50% as $\phi(b^2)$ increases. At base 3 (three odd characters) all products are positive. At base 23 (253 odd characters) the split is 129 to 124. The sign structure is not dominated by a majority.

b	odd chars	positive	negative	%pos	net
3	3	3	0	100	+0.62
5	10	8	2	80	+1.04
7	21	13	8	62	+1.81
11	55	27	28	49	+1.66
13	78	50	28	64	+3.45
17	136	74	62	54	+2.53
19	171	99	72	58	+5.11
23	253	129	124	51	+1.56
29	406	224	182	55	+5.32
31	465	249	216	54	+6.73
37	666	—	—	—	+8.25

TABLE 3. Sign split across prime bases at $s = 0.5$ (148,000 primes per base). The positive fraction drifts toward 50% as the number of characters grows. The net remains positive in every case.

Second, the net sum is positive at every base. The positive terms win not by number but by magnitude: the collision coefficients with the largest $|\hat{S}^\circ(\chi)|$ tend to pair with convergence-supporting (positive) products.

b	odd chars	top-1 sign	top-2 sign
3	3	+	+
5	10	+	+
7	21	+	+
11	55	+	+
13	78	+	+
17	136	+	+
19	171	+	+
23	253	+	+
29	406	−	−
31	465	+	+
37	666	+	+

TABLE 4. Sign of the product $\hat{S}^\circ(\chi) \cdot P(s, \chi)$ at $s = 0.5$ for the characters with the two largest $|\hat{S}^\circ|$. In 10 of 11 bases, the dominant coefficient supports convergence. Base 29 is the sole exception, but the net remains positive.

At base 29, the two largest coefficients are negative, but the third and fourth largest are positive with stronger phase alignment, producing a net contribution that overwhelms the negative leaders. The structural integrity does not depend on any single character.

6. The Anti-Correlation

The sign structure raises a question: why do the positive terms consistently outweigh the negative? The answer is an anti-correlation between the collision coefficients and the prime character sums.

b	odd chars	ρ (Pearson)	ρ_s (Spearman)
3	3	+0.00	−1.00
5	10	−0.45	+0.31
7	21	−0.28	+0.21
11	55	−0.23	−0.11
13	78	−0.27	−0.08
17	136	−0.32	−0.28
19	171	−0.13	−0.05
23	253	−0.26	−0.21
29	406	−0.24	−0.21
31	465	−0.24	−0.27
37	666	−0.27	−0.30

Table 5. Pearson and Spearman correlations between $|\hat{S}^\circ(\chi)|$ and $|P(0.5, \chi)|$ across odd characters. The Pearson correlation is negative at every base: the collision invariant places its largest Fourier weight on characters with moderate prime sums. 148,000 primes per base.

The Pearson correlation between $|\hat{S}^\circ(\chi)|$ and $|P(s, \chi)|$ is negative at every prime base tested, ranging from -0.13 to -0.45. Characters with large collision coefficients tend to have moderate prime character sums, and characters with large prime sums tend to have small collision coefficients.

The prime character sum $|P(s, \chi)|$ measures a character's sensitivity to the distribution of primes in residue classes and, through the relation $P = \log L - H$, to the zeros of the corresponding L-function. Characters with large $|P|$ are those most affected by nearby zeros.

The anti-correlation therefore suggests that the collision invariant places its largest Fourier weight on the characters *least* sensitive to L-function zeros. The digit geometry of $\lfloor br/p \rfloor$ appears to produce a

spectral profile that is misaligned with the zero-sensitive directions in character space.

This points to a structural mechanism behind the convergence of F°. The sum may converge not because the L-function zeros are absent, but because the collision invariant is not looking at them.

7. The Scramble Test

The anti-correlation appears to be load-bearing: in the scramble experiment, scrambling the assignment of collision coefficients to characters destroys the observed net.

Proposition 6. *Let π be a uniform random permutation of the odd characters. The expected value of the scrambled sum $\sum_\chi \hat{S}^\circ(\chi_\pi) \cdot P(s,\chi)$ is*

$$\mathbb{E}\Big[\sum_\chi \mathrm{Re}\big(\hat{S}^\circ(\chi_\pi) \cdot P(s,\chi)\big)\Big] = \frac{1}{n}\,\mathrm{Re}\Big(\sum_\chi \hat{S}^\circ(\chi) \cdot \sum_\chi P(s,\chi)\Big),$$

where n is the number of odd characters. By character orthogonality, $\sum_{\chi \text{ odd}} \chi(p) = (\phi(m)/2)(\delta_{p\equiv 1} - \delta_{p\equiv -1})$, so $\sum_\chi P(s,\chi)$ receives contributions only from primes $p \equiv \pm 1 \pmod{m}$.

Proof. For a uniform random permutation π,

$$\mathbb{E}[\hat{S}^\circ(\chi_{\pi(j)})] = \frac{1}{n}\sum_k \hat{S}^\circ(\chi_k)$$

for each j. Substituting and exchanging summation gives the displayed formula. The character sum identity is standard [2]. □

The z-scores are consistently positive and mostly above 2. A generic weighting of the same characters would produce a net near zero. The collision invariant's spectral profile is not merely different from random: in the tested bases, it is specifically oriented toward a positive net. The anti-correlation between $|\hat{S}^\circ|$ and $|P|$ is the leading candidate mechanism.

Remark 7. The scramble test is computational evidence, not a proof. Proving that the collision invariant's Fourier spectrum must be anti-correlated with the prime character sum magnitudes is an open problem. It would require understanding why the digit geometry of $\lfloor br/p \rfloor$ produces a spectral profile transverse to the zero-sensitive directions in character space.

b	odd	actual	$\mathbb{E}$[scr]	σ[scr]	z
7	21	+1.81	−0.16	0.92	+2.1
11	55	+1.66	−0.06	1.27	+1.4
13	78	+3.45	−0.05	1.43	+2.5
17	136	+2.53	−0.05	1.64	+1.6
19	171	+5.11	−0.07	1.66	+3.1
23	253	+1.56	−0.00	1.87	+0.8
29	406	+5.32	+0.02	2.09	+2.5
31	465	+6.73	−0.00	2.27	+3.0

TABLE 6. The actual net $F^\circ(0.5)$ compared with 10,000 random permutations of the collision coefficients. Proposition 6 gives the exact expected scrambled mean; in the tested bases this mean is near zero. The actual net is 1–3 standard deviations above, indicating the specific coefficient-to-character assignment is relevant to the observed convergence pattern.

8. Consequences

Remark 8. The mixed-sign data shows that any implication from convergence of F° at $s = 1/2$ to individual zero-freeness of every $P(s, \chi)$ cannot proceed by a positivity argument. If convergence of F° at $s = 1/2$ holds, it is compatible with cancellation between terms of opposite sign.

The collision transform is therefore not a detector of individual L-function zeros. If F° converges at $s = 1/2$, the statement is not that every odd L-function modulo m is zero-free for $\mathrm{Re}(s) > 1/2$. The statement is that the zeros, if any, are arranged so that their effects cancel under the collision weighting. This is a constraint on the geometry of the joint zero set, not on its emptiness.

Remark 9. The ratio of positive to negative total magnitude is approximately 2:1 at $s = 0.5$ in base 10 (Table 2). Whether this ratio is universal, base-dependent, or depth-dependent is an open question.

9. Remarks

What is proved. The general neutrality theorem is proved for all odd primes q not dividing b, using only the reflection identity and the invertibility of 2 modulo q. Computational verification covers 117 cases (all bases $b \leq 20$, primes $q \leq 19$).

The sign structure, anti-correlation, and scramble behavior are computational observations supported by 11 prime bases with up to 666 odd characters each. They are not theorems.

The structure at $q = 2$. The prime $q = 2$ is excluded because 2 is not invertible modulo 2. The parity structure of the collision invariant depends on the parity of b and requires separate analysis.

The arc. The neutrality theorem at $q = 3$ controls the cancellation mechanism that allows F° to penetrate below $s = 1$ [4]. The general neutrality theorem shows this is the first instance of a family of constraints at every prime scale. The computed sign structure indicates that the convergence of F° is not a GRH equivalence but a collective phenomenon: the collision invariant creates a structured weighting of L-functions in which individual zeros can be absorbed by the cancellation between terms of opposite sign.

References

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